

Affiliated Event Proposal

Mathematics for Post-Quantum Cryptography

(MathPQC 2026)

Proposal for ASIACRYPT 2026

Conference: ASIACRYPT 2026, December 7-11, 2026, Hong Kong, China

Affiliated-event day and venue: Monday, December 7, 2026, Hong Kong Polytechnic University

1. Name (and possible acronym) of the event

Mathematics for Post-Quantum Cryptography (MathPQC 2026)

2. Names and contact information of the organizers

Organizer 1 (onsite): Shi Bai, Shanghai Jiao Tong University, baishi@sjtu.edu.cn

Organizer 2 (onsite): Mingjie Chen, COSIC, KU Leuven, mjchen555@gmail.com

Organizer 3 (onsite): Steven Galbraith, University of Auckland, S.Galbraith@auckland.ac.nz

3. Length of event (half or whole day)

Whole day.

4. Proposed format and potential speakers

The proposed format is a one-day workshop featuring invited talks and contributed talks.

Potential speakers

John Voight, Peter Kutas, San Ling, Hiroshi Onuki, Elena Kirshanova, Jung Hee Cheon etc

5. Abstract summarising the proposed event and its justification

Post-quantum schemes rely on several families of computationally hard problems, including lattice problems (e.g. the shortest vector problem on Euclidean and ideal lattices, and the lattice isomorphism problem on Euclidean and module lattices), finding isogenies between supersingular elliptic curves (and superspecial abelian varieties) over finite fields, decoding problems for linear codes, and solving systems of multivariate quadratic (MQ) equations over finite fields. These problems draw on a broad range of mathematical disciplines, including number theory, algebra, the geometry of numbers, algebraic geometry, and many others.

In light of these connections, this affiliated event aims to bring together mathematicians working in these fields and cryptographers interested in the mathematical foundations of post-quantum cryptography. The event will provide an opportunity to exchange ideas across disciplines, highlight recent developments, and foster new interactions between the mathematics and cryptography communities.

6. In the case of a submission process

Not applicable.

7. Estimated number of attendees total and expected number of student attendees

Estimated attendance: 30 to 50 participants in total.

Estimated students attendance: 20 to 30 participants in total.

8. List of specific potential attendees

Steven Galbraith, Mingjie Chen, Shi Bai, Alain Couvreur, Thomas Debris-Alazard, Jintai Ding, Luca De Feo, Péter Kutas, Chloe Martindale, Edoardo Persichetti, Ling Song, André Schrottenloher, Damien Stehlé, Yixin Shen, Wessel van Woerden, Bo-Yin Yang, Tanja Lange, Jean-Pierre Tillich, John Voight, Elena Kirshanova, Jung Hee Cheon, Chenfeng He, Yi-Fu Lai, Kaizhan Lin, Zhuo Huang, Zhi Hu, Xuzhe Liu, Yunqi Wen, Zheng Xu, Ling Qin, Peigen Li, Tomoki Moriya etc

9. Biographical sketch for each organizer

Organizer 1 bio: Shi Bai is a Professor at Shanghai Jiao Tong University. His research interests are in lattice-based cryptography. He received his Ph.D. in Computer Science from the Australian National University. Before joining Shanghai Jiao Tong University, he was a postdoctoral researcher at the University of Auckland and École Normale Supérieure de Lyon, and later served as an Assistant and Associate Professor at Florida Atlantic University.

Organizer 2 bio: Mingjie Chen is a postdoctoral researcher at COSIC, KU Leuven, working on post-quantum cryptography, particularly isogeny-based cryptography. She received her Ph.D. in number theory from UC San Diego in 2022. Before joining KU Leuven, she was a postdoctoral researcher at Université Libre de Bruxelles from February 2024 to June 2024 and at the University of Birmingham from March 2022 to January 2024.

Organizer 3 bio: Steven Galbraith is a Professor of Mathematics at the University of Auckland. His research is in computational number theory, computational algebraic geometry and applications in public key cryptography. Currently his main research focus is post-quantum cryptography. He has supervised more than 15 PhD students, published over 100 papers, chaired several major conferences (including ASIACRYPT), and served on more than 50 conference program committees.

10. Funding sources that may be used and for what purposes

No external funding is currently planned for the event. The organizers expect to cover their own travel costs for attending ASIACRYPT.

11. Information about previous similar events

ASIACRYPT 2026 affiliated event “LatticeCC: Workshop on Lattice Coding and Crypto” had approx 60 participants, and included several talks on the mathematical background of post-quantum cryptography.

12. Link to draft website

Not available at this time. We will provide one later.